Parkinson's Disease Progression and Treatment Dynamics Accounting for Nonlocality of Bioneurological Processes

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Abstract. Parkinson's disease (PD) is a complicated neurodegenerative condition marked by the gradual degradation of dopaminergic neurons in the substantia nigra, which causes motor and non-motor symptoms. Understanding the dynamics of Parkinson's disease progression and evaluating treatment techniques necessitates a complete assessment of the nonlocal character of the bioneurological mechanisms involved. This research explores a novel approach to model PD progression by integrating nonlocality into analysing bioneurological processes. Nonlocality in this context refers to the interdependence of the disease progression as the long-term memory. In addition, we have investigated the impact of the two therapeutic interventions on the dynamics of PD, the reduction of extracellular α-synuclein and controlling the activated microglia. Our findings suggest that the memory effect delays the onset of Parkinson's disease.

Keywords: Parkinson's disease · microglia · alpha-synuclein

1 Introduction

Parkinson's disease (PD) is a neurological condition that mostly impacts motor function [1]. It is typified by a gradual loss of dopaminergic neurons in the spinal cord, a part of the brain that is important for motor control and other processes. It is thought to be caused by a mix of environmental and genetic factors, while its exact cause is yet unknown. PD is diagnosed mostly by clinical assessment, which includes a full medical history and a thorough neurological examination. There is no conclusive test for PD; thus, physicians rely on the presence of specific symptoms and the elimination of other possible causes [2]. Ongoing research is being conducted to investigate new disease-modifying medications and get a

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better knowledge of PD processes. Recent advances in genetics, neuroimaging, and molecular biology have offered important insights into the disease's complexity [3].

The pathological characteristic of Parkinson's disease is the creation of aberrant protein deposits known as Lewy bodies within the brain's nerve cells [4]. These Lewy bodies predominantly comprise aggregated alpha-synuclein (α-syn), a protein that typically regulates neurotransmitter release. α-synuclein buildup disturbs normal cellular function and causes the death of dopamine-producing neurons in the brain's substantia nigra [5]. Research is being conducted to understand better the specific molecular mechanisms that cause α-syn aggregation and its function in Parkinson's disease development. Therapeutic techniques targeting α-syn, including strategies to prevent its aggregation or improve its clearance, are currently being investigated as possible therapies for Parkinson's disease.

On the other hand, microglia play an important role in the immune response of the central nervous system (CNS) and have been linked to the development of neurodegenerative illnesses such as Parkinson's disease [6,7]. Microglia are the CNS's resident immune cells, responsible for monitoring their surroundings, maintaining homeostasis, and reacting to damage or infection [8]. In the setting of Parkinson's disease, microglia are hypothesized to contribute to the neuroinflammatory processes that accompany disease development [9]. They can exist in different states, including an activated or resting state, and their activation status is crucial in the context of neuroinflammation and neurodegeneration.

Activated microglia are involved in the immune response and can release pro-inflammatory molecules, such as cytokines and reactive oxygen species [10]. In Parkinson's disease, there is evidence of chronic neuroinflammation, and activated microglia are often found in the affected brain regions, particularly the substantia nigra. The substantia nigra is a brain region that undergoes degeneration in Parkinson's disease, leading to a decrease in dopamine production [11]. The exact role of activated microglia in Parkinson's disease is complex and not fully understood. On the one hand, they can contribute to the clearance of damaged cells and debris, but on the other hand, their prolonged activation can lead to a harmful, chronic inflammatory environment that contributes to the death of dopaminergic neurons [12].

Resting microglia, in contrast, are considered more quiescent and less inflammatory. They are involved in maintaining tissue homeostasis and surveilling the microenvironment. In Parkinson's disease, there is evidence that the transition from resting to activated microglia is a dynamic process that occurs in response to various signals, including the presence of misfolded proteins like α-synuclein, a hallmark of Parkinson's disease [9].

Research is ongoing to better understand the role of microglia in Parkinson's disease and to develop therapeutic strategies that modulate their activation state. Some potential approaches have been applied for a deeper understanding and the development of effective treatments, including target-specific signalling pathways involved in microglial activation or developing drugs that promote the resolution of inflammation [13].

In this work, we focus on the dynamics of five substances, namely, neurons, infected neurons, α-synuclein, and resting and activated microglia. There are different modelling approaches to see the dynamics of the substances, such as the traditional ordinary differential equation (ODE), fractional differential equation (FDE), partial differential equation, etc. In general, the behaviour of a system at a given point depends on the history of that system and the traditional ODE approach can not address such scenarios [14,15]. Therefore, the FDE approach is needed in studying PD because the disease develops in the brain slowly and affects over the long term. Furthermore, fractional derivatives introduce additional parameters, such as the fractional order, allowing for greater flexibility in modelling systems with complex dynamics. Here, we apply ODE and FDE approaches to substance interactions and compare the disease progressions in the brain. The fractional modelling approach is essential in studying PD as it develops slowly in the brain, and the disease's current status depends on its history. This advantage makes the study of fractional order systems an active area of research, which accounts for the nonlocality dynamics of neural networks in the brain. Understanding the fractional order dynamics of brain circuits gives more insight into the changes observed in Parkinson's disease.

The rest of this paper is organized as follows. In Sect. 2, we describe the ordinary and fractional differential equation models followed by linear stability analysis for the equilibrium points in Sect. 3. Simulation results are presented in Sect. 4 followed by summary in Sect. 5.

2 Models for the Analysis of PD Dynamics Accounting for Nonlocality of Neurobiological Processes

For $t > 0$, suppose $N(t)$ is the density of healthy brain neurons; $I(t)$ is the density of infected neurons; $S(t)$ is the density of extracellular α-synuclein (α-syn); $M_r(t)$ and $M_a(t)$ are the densities of rested and activated microglia, respectively. Following [16], we consider the model given by a system of nonlinear ordinary differential equations as:

$$\frac{dN}{dt} = \tau_1 - \beta(1-\sigma_1)NS - \mu_1 N - a(1-\sigma_2)NM_a, \tag{1a}$$

$$\frac{dI}{dt} = \beta(1-\sigma_1)NS - d_1 I - a(1-\sigma_2)IM_a, \tag{1b}$$

$$\frac{dS}{dt} = ed_1 I - a(1-\sigma_2)SM_a, \tag{1c}$$

$$\frac{dM_a}{dt} = a(1-\sigma_2)(I+S)M_r - \mu_2 M_a, \tag{1d}$$

$$\frac{dM_r}{dt} = \tau_2 - a(1-\sigma_2)(I+S)M_r + \mu_2 M_a - \mu_3 M_r, \tag{1e}$$

with non-negative initial conditions. Here, the parameter τ_1 is the rate of neuron formation due to neurogenesis in the substantianigra and striatum, and τ_2

is the microglia proliferate rate. A neuron becomes infected when extracellular α-syn behaves like a prion and spreads into healthy neuron cells at a rate of β. An infection-related neuron dies at a rate of d_1 due to α-syn aggregation. In the extracellular area, α-syn is released in proportion e upon the lysis of infected neurons. For Parkinson's disease, the goal of treatment is to either decrease extracellular α-syn's absorption or reduce CNS inflammation brought on by activated microglia. The parameter σ_1 represents the effective ratio of the therapeutic drug in overcoming extracellular α-syn, and σ_2 is the effective ratio for controlling activated microglia with the rate a. The parameter μ_2 represents the repressed rate of activated microglia to return to a resting state when the stimulation has subsided. Additionally, neurons and microglia undergo apoptosis at rates of μ_1 and μ_3, respectively. Here, all the parameters involved in the equations are taken to be positive based on their corresponding biomedical interpretation.

In modern neuron modelling, fractional-order differential equations have been shown to be the best choice for comprehending the dynamic behaviour underlying the experimental data obtained by neurobiologists. Examples include the tissue-electrode interface [17], the pharmacokinetics of drug administration and absorption [18,19], the mechanical characteristics of viscoelastic tissue [20], and anomalous calcium sub-diffusion in micro-domains [21]. Biological processes often exhibit memory, where the current state depends not only on the immediate past but also on a more extended history. Neurons adapt with time scale dynamics, and the fractional differential equation captures such time scale behaviour, commonly known as long-term memory behaviour [22]. Here, we introduce such long-term memory behaviour in the model (1) as:

$$D_t^\eta N = \tau_1 - \beta(1-\sigma_1)NS - \mu_1 N - a(1-\sigma_2)NM_a, \tag{2a}$$

$$D_t^\eta I = \beta(1-\sigma_1)NS - d_1 I - a(1-\sigma_2)IM_a, \tag{2b}$$

$$D_t^\eta S = ed_1 I - a(1-\sigma_2)SM_a, \tag{2c}$$

$$D_t^\eta M_a = a(1-\sigma_2)(I+S)M_r - \mu_2 M_a, \tag{2d}$$

$$D_t^\eta M_r = \tau_2 - a(1-\sigma_2)(I+S)M_r + \mu_2 M_a - \mu_3 M_r, \tag{2e}$$

with non-negative initial conditions. The notation $D_t^\eta(\cdot)$ denotes the Caputo fractional derivative and it is defined by

$$D_t^\eta Y = \frac{1}{\Gamma(1-\eta)} \int_0^t \frac{Y'(s)}{(t-s)^\eta} ds, \quad 0 < \eta < 1,$$

where Γ is the gamma function. The novelty of this fractional differential equation (FDE) is that it takes care of different hereditary behaviours of biological processes, including non-local aspects of phenomena that are prevalent in various biological systems, including the brain. This fractional order in FDEs serves as an extra parameter that can be adjusted to capture different degrees of memory and nonlocality in biological processes.

3 Stability of Disease Progression Dynamics with Traditional and Fractional Models

Stability theory is an important feature in dynamic problems involving disease development and therapy. It serves as a foundation for predictability, effective treatment planning, and therapeutic strategy optimization, all of which contribute to better results for those suffering from neurological illnesses and other dynamic health difficulties. In addition, stability analysis provides insights into the robustness of the system against perturbations.

Linear stability analysis is a mathematical approach that examines the stability of equilibrium points in ordinary differential equation (ODE) models. It entails linearizing the nonlinear ODE system around an equilibrium point and determining the eigenvalues of the resultant linear system. The eigenvalues indicate whether disturbances from the equilibrium point increase or decrease. The equilibrium points of the model (1) are the steady-states of it and they are the solution of the system of algebraic equations:

$$\tau_1 - \beta(1-\sigma_1)NS - \mu_1 N - a(1-\sigma_2)NM_a = 0, \tag{3a}$$
$$\beta(1-\sigma_1)NS - d_1 I - a(1-\sigma_2)IM_a = 0, \tag{3b}$$
$$ed_1 I - a(1-\sigma_2)SM_a = 0, \tag{3c}$$
$$a(1-\sigma_2)(I+S)M_r - \mu_2 M_a = 0, \tag{3d}$$
$$\tau_2 - a(1-\sigma_2)(I+S)M_r + \mu_2 M_a - \mu_3 M_r = 0. \tag{3e}$$

This system has at most five solutions: one can be determined explicitly as $E_0 = (\tau_1/\mu_1, 0, 0, 0, \tau_2/\mu_3)$, and the others depend on the number of real roots of the fourth degree polynomial equation

$$q_4 x^4 + q_3 x^3 + q_2 x^2 + q_1 x + q_0 = 0, \tag{4}$$

where $q_4 = \tau_2 a^5 (1-\sigma_2)^5/\mu_3$, $q_3 = \beta(1-\sigma_1)ed_1\mu_2 a^2(1-\sigma_2)^2 + \tau_2 a^4(\mu_1 + d_1 + ed_1)(1-\sigma_2)^4/\mu_3$, $q_2 = \beta(1-\sigma_1)ed_1^2\mu_2 a(1-\sigma_2) + \tau_2 a^3 d_1(\mu_1 + e\mu_1 + ed_1)(1-\sigma_2)^3/\mu_3$, $q_1 = \tau_2 a^2 ed_1(1-\sigma_2)^2(d_1\mu_1 - \tau_1\beta(1-\sigma_1))/\mu_3$, and $q_0 = -\tau_1\tau_2 a\beta e^2 d_1^2(1-\sigma_1)(1-\sigma_2)/\mu_3$. In this case, the component M_a^* of a non-trivial equilibrium point E_* corresponding to activated microglia satisfies the polynomial equation. The components for N, I, S, and M_r are given by $N^* = (a(1-\sigma_2)M_a^*(d_1 + a(1-\sigma_2)M_a^*))/(e\beta d_1(1-\sigma_1))$, $I^* = a(1-\sigma_2)S^*M_a^*/(ed_1)$, $S_* = ed_1\mu_2 M_a^*/(a(1-\sigma_2)M_r^*(a(1-\sigma_2)M_a^* + ed_1))$ and $M_r^* = \tau_2/\mu_3$.

In the polynomial equation (4), the coefficients q_4, q_3, and q_2 are always positive, q_0 is negative, and q_1 may change its sign. Descartes' rule of signs states that the number of positive roots is equal to the changes in coefficients' signs of the polynomial equation or less than that number by an even integer [23]. Therefore, irrespective of the sign change of q_1, the polynomial equation has exactly one sign change of the coefficients. Hence, the equation (4) has at least one positive root M_a^*, which is used in finding the other components for the non-trivial equilibrium point. In addition, the systems (1) and (2) have a unique non-trivial equilibrium point for the parameter values given in Table 1.

The stability behaviours of the equilibrium points for the non-fractional and fractional models depend on the signs of the eigenvalues corresponding to the Jacobian matrix. For the non-fractional model (1), if real parts of all the eigenvalues are negative, then the equilibrium point is stable; otherwise, it is unstable. On the other hand, an equilibrium point for the fractional model (2) is stable if all the eigenvalues (say λ_i) of the Jacobian matrix satisfy $|\arg(\lambda_i)| > \eta\pi/2$, for a fixed η. The equilibrium point E_0 is always unstable for the non-fractional and fractional models, and the non-trivial equilibrium point E_* is stable. This stability analysis offers insights into the stability of various states in the neural system affected by PD. In addition, it helps in understanding how changes in the parameters of the system influence the stability behaviour of the system. Furthermore, it guides the development of interventions that aim to restore or maintain a stable neural environment in PD.

Table 1. Temporal parameter values [16].

Parameter	Value	Unit
τ_1	10^{-4}	g/ml/day
β	0.5	ml/g/day
σ_1	10^{-2}	$\cdot$
μ_1	1.9×10^{-4}	day^{-1}
a	0.02	ml/g/day
σ_2	10^{-2}	$\cdot$
d_1	3.4×10^{-4}	day^{-1}
e	0.5	$\cdot$
μ_2	6×10^{-3}	day^{-1}
τ_2	5×10^{-4}	g/ml/day
μ_3	1.5×10^{-2}	day^{-1}

4 Results and Discussions

The developed mathematical model incorporates the dynamics of Parkinson's disease progression and treatment effects, along with its numerical implementation. In this section, we analyze and simulate the models (1) and (2) numerically, which allows for the systematic exploration of various optimal settings that maximize therapeutic efficacy, such as the long-term memory behaviour, the reduction of extracellular α-synuclein, and controlling the activated microglia. We have adopted the temporal parameter values involved in the models from [16], which are given in Table 1. Additionally, in-house tools based on Matlab and C-language have been used for the simulations. Nevertheless, the codes will be provided as a supplement to the extension of this work to a journal paper.

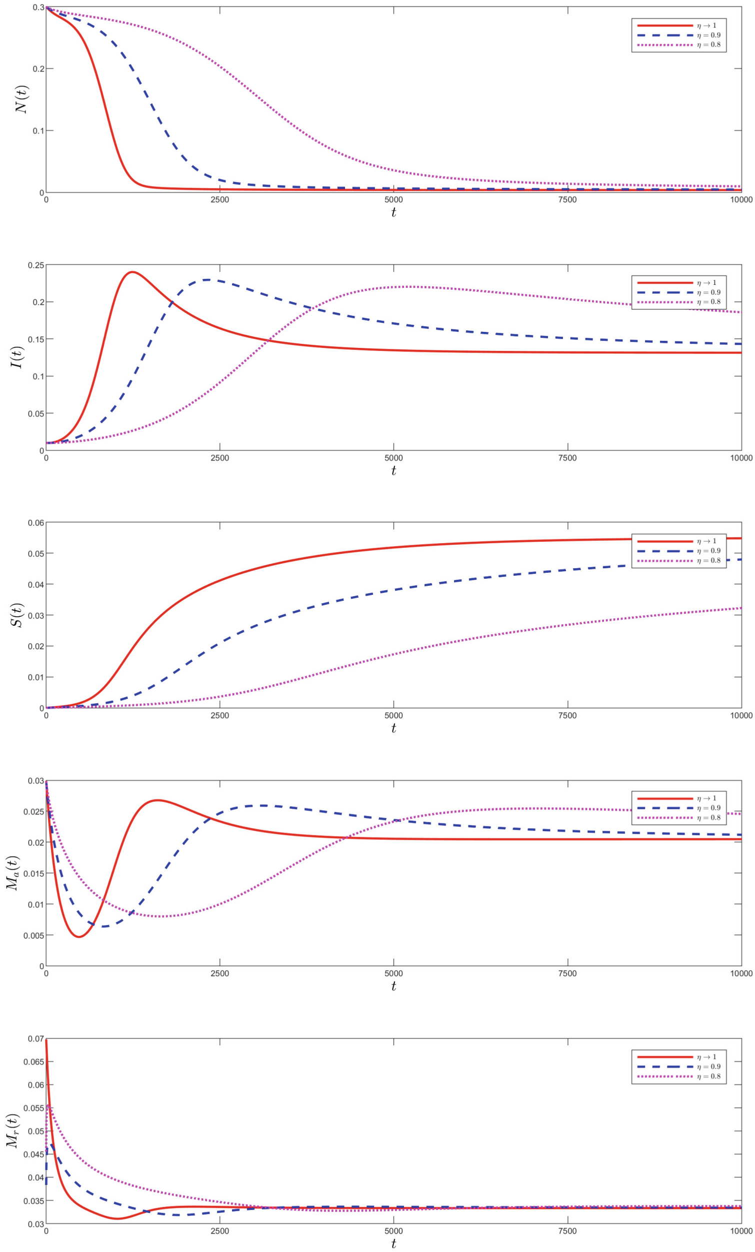

Fig. 1. (Color online) Solutions for the non-fractional and fractional models for a fixed initial condition.

The non-trivial equilibrium point E_* is stable for both non-fractional and fractional models for the parameter values given in Table 1. We have solved both the models numerically with the initial condition $(N(0), I(0), S(0), M_a(0),$

$M_r(0)) = (0.3, 0.01, 0.00001, 0.03, 0.07)$, and their simulation results are presented in Fig. 1. This shows that a small change in the early stages does not have a significant impact on the later stages. Here, both the models converge to the non-trivial equilibrium point E_*, but their convergence rate differs. Figure 1 shows that the non-fractional model converges to the equilibrium point E_* in the considered time range $[0, 10000]$. But, the fractional model does not converge in the considered time range for $\eta = 0.9$ as we see the evolution for infected neurons still not converge to the corresponding equilibrium component $I^* = 0.133$ at $t = 10000$.

For the non-fractional model, Fig. 1 shows that the density of the healthy neurons slowly decreases for the chosen initial condition. In addition, the density of the infected neurons shoots up to a maximum density, then slowly decreases and resting at I^*. The same behaviour is observed for the activated microglia. Furthermore, the resting microglia rapidly decrease then saturated to M_r^*. On the other hand, the density of the healthy neurons in the fractional model decreases more slowly than in the non-fractional model. In addition, the maximum shoot-up densities for the infected neurons and activated microglia are less for the fractional model than for the non-fractional model. With a decrease value of η, these maximum values reduce significantly, as shown in Fig. 1.

The infection rate of the healthy neuron cells also plays a crucial role in the progression of Parkinson's disease. With an increase in the value of β, the density of the healthy neurons decreases, and the density of the infected neurons increases. Figure 2 depicts the progression of activated and resting microglia for different values of β. It shows that the equilibrium components for the activated and resting microglia are independent of β, but their progression rate varies.

Next, we move to the cases where some PD treatments can be applied. In the considered models, the parameter σ_1 is involved in the treatment to reduce

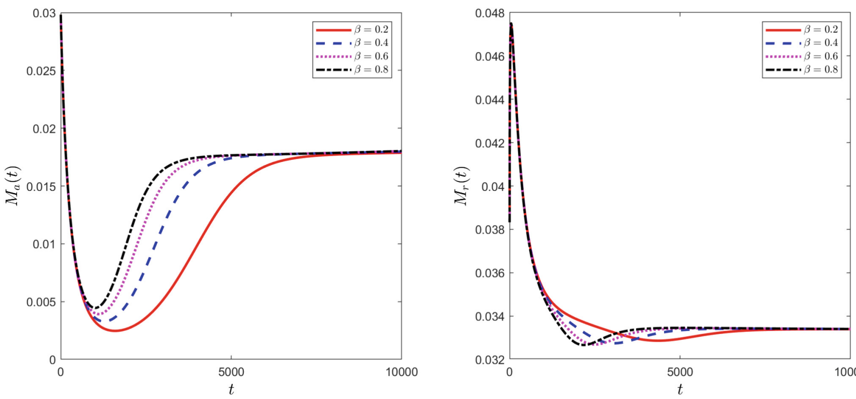

Fig. 2. (Color online) Effects of the parameter β on activated and resting microglia for the fractional model with $\eta = 0.9$.

extracellular α-syn, and the parameter σ_2 is involved in inhibiting the activity of the microglia. Here, the parameters $\sigma_1 = 0$ and $\sigma_2 = 0$ are corresponding to no-treatments. First, we consider only the treatment related to extracellular α-syn, i.e., σ_1, and its results are plotted in Fig. 3. In this case, both models predicted that the density of healthy neurons increases with σ_1 increases, which forces in delaying the infection of neurons, and fewer neurons are infected. Moreover, the fractional model with $\eta = 0.9$ helps in more time delay to the infection of the neurons compared to the non-fractional model. Furthermore, it helps to increase the number of healthy neurons compared to the no-treatment case. Although changing equations to highlight various parts of α-synuclein control that may be targeted is conceptually simple, it requires significant technical efforts. Once possible modifiable locations have been discovered, researchers can develop medications to target these areas. This approach entails screening compounds, frequently using in vitro and in vivo tests, to identify molecules that may successfully interact with the target.

Figure 4 shows the comparisons between the non-fractional and fractional models for different values of the treatment ratio parameter σ_2. In both cases,

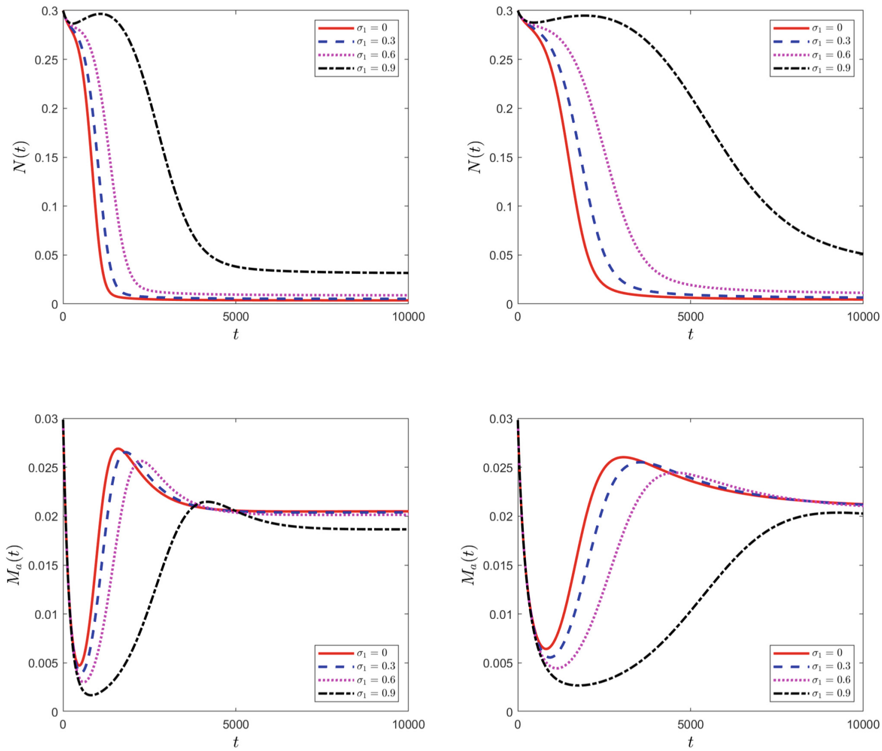

Fig. 3. (Color online) Solutions for the non-fractional (left panel) and fractional (right panel) models with $\eta = 0.9$ for different values of the treatment ratio parameter σ_1.

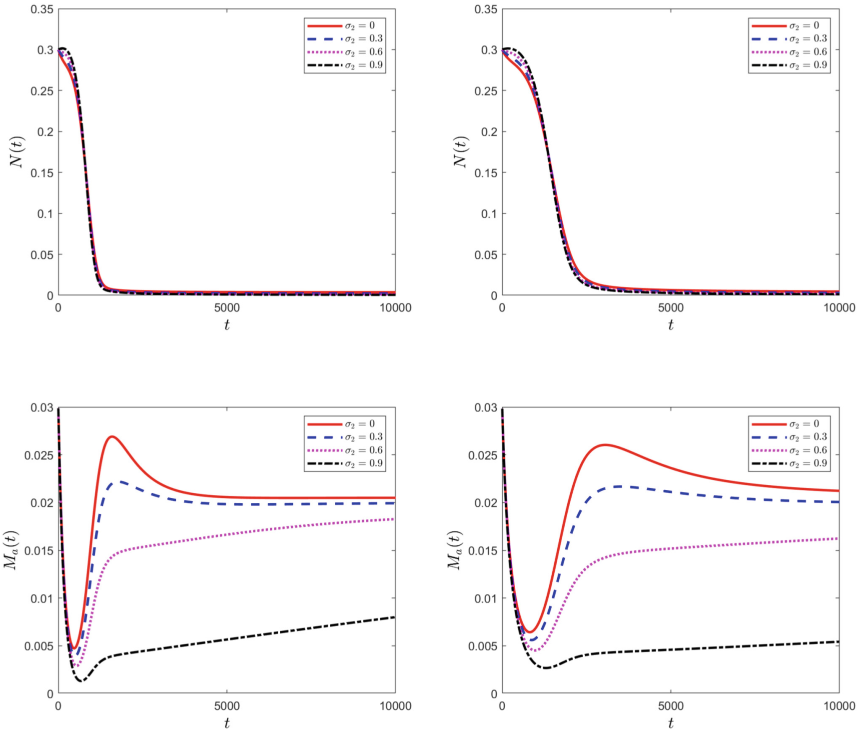

Fig. 4. (Color online) Solutions for the non-fractional (left panel) and fractional (right panel) models with $\eta = 0.9$ for different values of the treatment ratio parameter σ_2.

there is no significant increase in the density of the healthy neurons. However, the density of the activated microglia decreases with an increase of the parameter σ_2, which gives a faster extracellular α-syn accumulation and a fast degeneration in the healthy neurons.

Finally, we mix both therapeutic modalities and investigate their effects on Parkinson's disease development. The settings of the treatment parameters are chosen to represent four cases: (i) low ratio in both treatments ($\sigma_1 = \sigma_2 = 0.3$), (ii) high ratio in the first treatment ($\sigma_1 = 0.6, \sigma_2 = 0.3$), (iii) high ratio in the second treatment ($\sigma_1 = 0.3, \sigma_2 = 0.6$), and (iv) high ratio in both treatments ($\sigma_1 = \sigma_2 = 0.6$). We plot the results for the fractional model ($\eta = 0.9$) in Fig. 5 for all of these cases with the no-treatment case. A significant delay occurred in healthy neurons for cases (iii) and (iv); however, there was a decrease in activated microglia. Overall, these treatments delay the onset of PD.

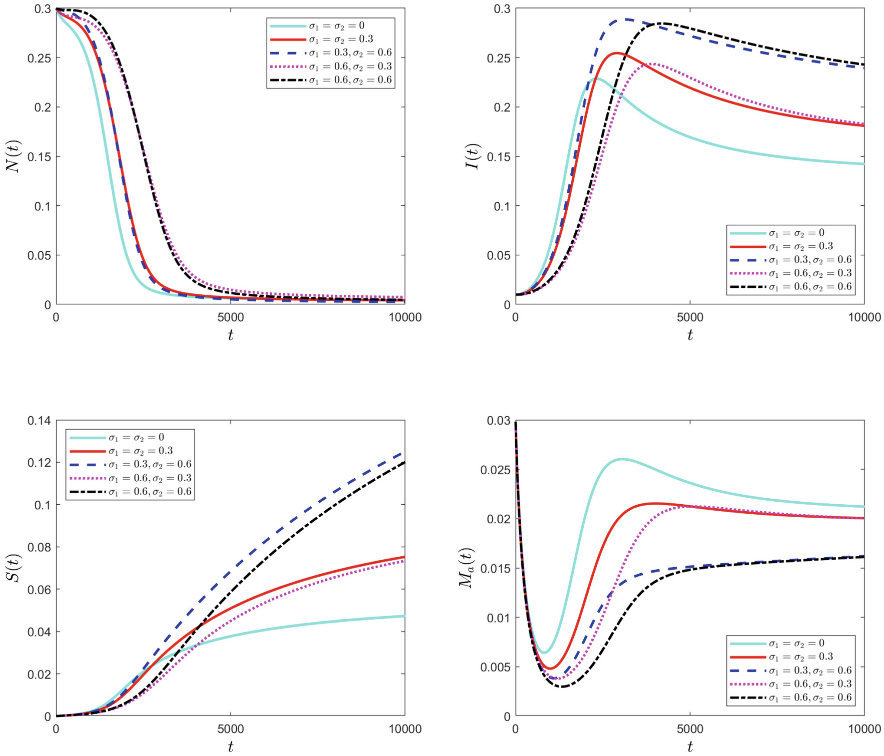

Fig. 5. (Color online) Solutions for the fractional model with $\eta = 0.9$ for different values of treatment ratios parameters σ_1 and σ_2.

5 Conclusions

In this paper, we have introduced a fractional differential equation model which accounts for the dynamics of healthy and infected neurons along with α-syn and microglia. This modelling approach has allowed us to study nonlocality as the long-term memory effect, which can not be observed through the traditional ODE model used in [9].

The numerical simulations suggest that the initial condition does not affect the onset of PD. On the other hand, the parameter involved in the fractional order η plays a crucial role in the speed of the disease progression. A decrease in the parameter η slows the speed of the disease progression. We have shown that the lower infection rate of the healthy neurons takes a longer time to converge to the disease state, which indicates the role of infection rate on PD. Therefore, the infection rate can control the speed towards PD. Furthermore, we have compared the results for the therapeutic interventions of extracellular α-syn and activated microglia. The simulation results show that the therapeutic interventions of reducing extracellular α-syn is more effective than inhibiting the activity of the microglia.

Our main goal here has been to study the effect of long-term memory on PD. So, as a starting point in this direction, we have considered the quadratic nonlinearity. As a next step, we shall proceed to systematic comparisons and validation of the model predictions with clinical data and the evaluation of the model's predictive power and reliability. While the experimental results are currently unavailable for this particular setting, other types of nonlinearities, such as Holling type II and other functional responses, could be implemented within the computational framework developed in this study. In the meantime, based on the practically plausible nonlinearity setting, we have demonstrated that the memory effect accounted for in this model for the first time delays the onset of Parkinson's disease.

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